Advanced Machine learning Mastering Course

Introduced by

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Agenda

1 Supervised Learning

2 Introduction to Regression

3 More Regression

4 Regression in Sklearn

5 Gradient decent-Normal equation

6 Regularization

7 Logistic Regression

8 Decision Tree

10 Neural Networks

11 Neural Nets Mini-Project

11 Math behind SVMs

11 SVMs in Practice

12 Instance Based Learning

13 Regression in Sklearn

14 Naive Bayes

15 Bayesian Inference

16 Ensemble B&B

17 Finding donors for CharityML

Boosting Algorithms:
AdaBoost

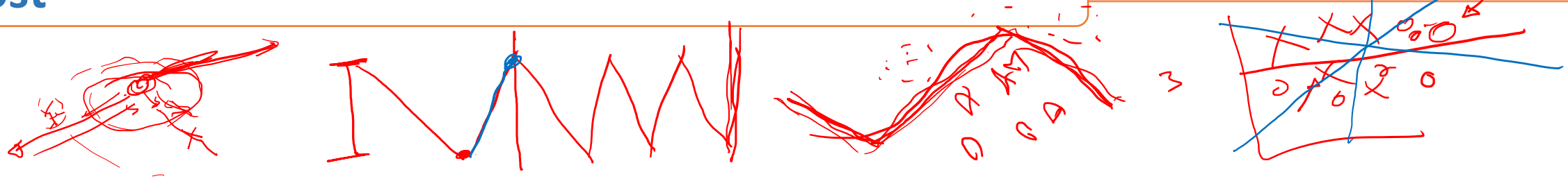
AdaBoost

Iterations

For $i = 1, \ldots, T$ or until low enough error is achieved:

- Fit weak learner $f_t(x)$ to data points with data point weights $\boldsymbol{\alpha}_i$
- Compute weak learner's weight $\boldsymbol{w_t} = \frac{1}{2}\ln\left(\frac{1 - weighted_error(f_t)}{weighted_error(f_t)}\right)$
- Recompute data point weights $\boldsymbol{\alpha}_i = \begin{cases} \alpha_i e^{-w_t} & if\ f_t(x_i) = y_i \\ \alpha_i e^{w_t} & if\ f_t(x_i) \neq y_i \end{cases}$
- Normalize weights $\boldsymbol{\alpha}_i$

Boost



Initialize weights to all training points

$$w = \frac{1}{N}$$

$w, b = 0$

$\frac{1}{3}$

Calculate error rate for each weak classifier

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Classifier with the lowest error rate

Now compute Voting Power for the classifier

$$\alpha = \frac{1}{2} \log \frac{1-\epsilon}{\epsilon}$$

Append the classifier in the ensemble classifier and heck if the classifier is good enough?

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

Update weights of each point where the previous classifier went wrong

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} & \text{if point classified correctly} \\ \frac{w_{old}}{2\epsilon} & \text{if point classified wrongly} \end{cases}$$

Boost

classification: step by step

1. Initialize the weight 1/m for all i=1..m examples

2. We create a decision stump for each variable and see how well each stump classifies samples to their target classes. We'd look at how many samples are correctly or incorrectly classified as Fit or Unfit for each individual stump.

3. We calculate the Gini index and build the tree

4. Total error for a stump is the sum of the weights associated with the incorrectly classified samples

5. We use the total error to determine the Amount of Say.
   Amount of say =1/2 log (1-total error/total error)

6. We need to modify the weights by using the below formula
   New Sample Weight = Sample Weight * e^amount of say

7. More weight is assigned to the incorrectly classified samples so that they're classified correctly in the next decision stump. Weight is also assigned to each classifier based on the accuracy of the classifier, which means high accuracy = high weight!

8. Reiterate from Step 2 until all the data points have been correctly classified, or the maximum iteration level has been reached

Boost

$$w = \frac{1}{N}$$

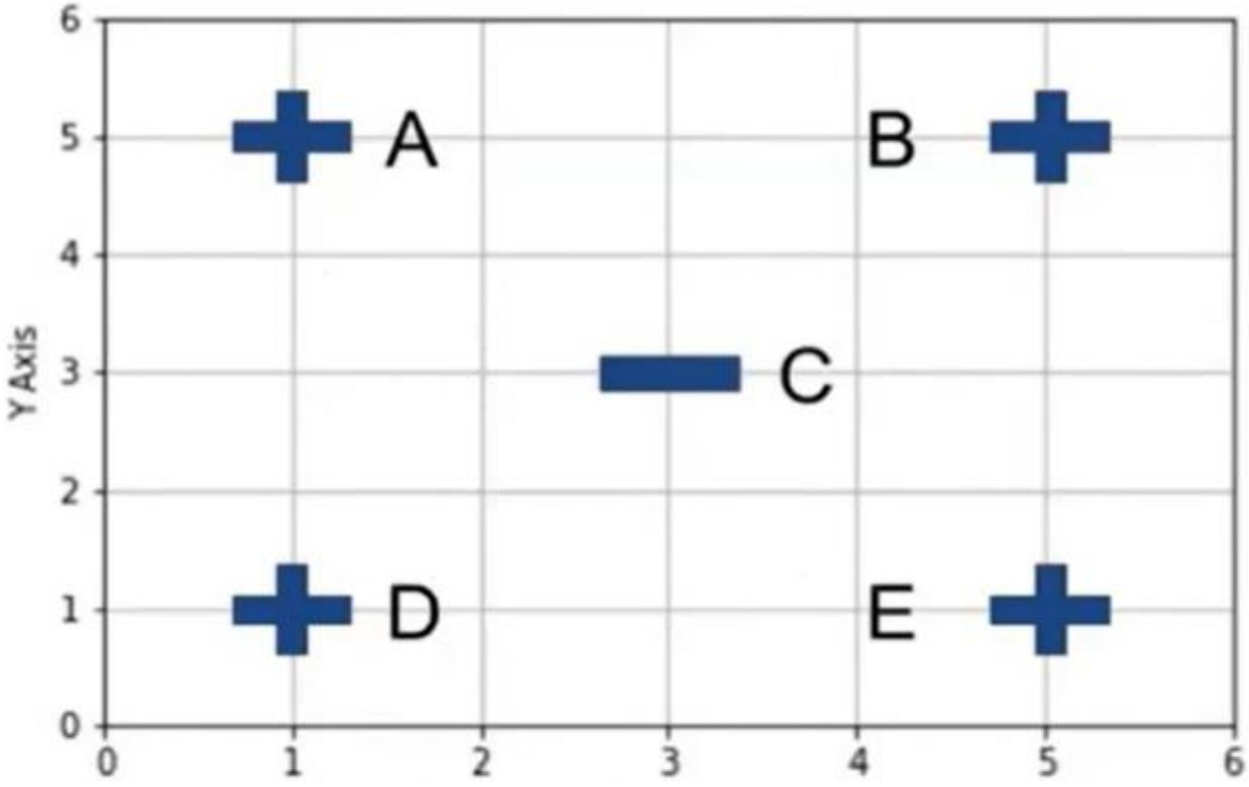
$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

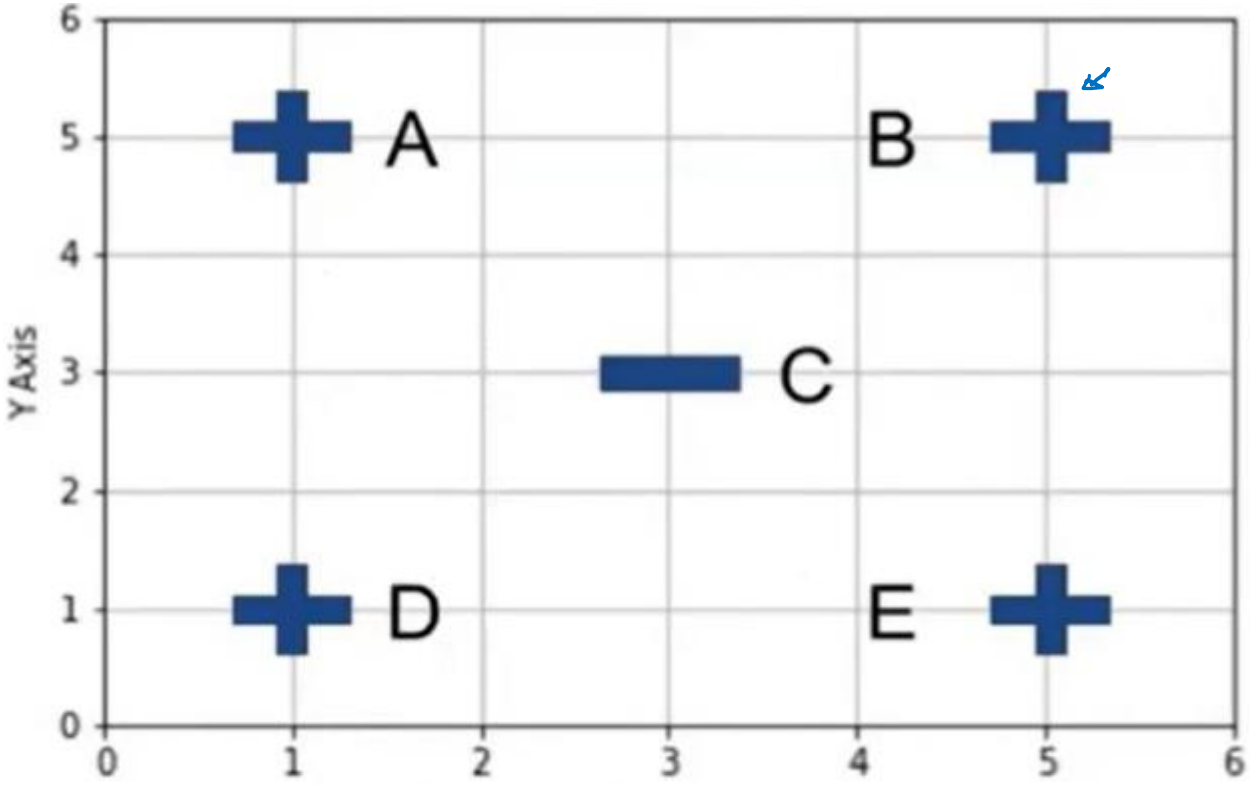
$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$



- Let's start with 5 points.
- 4 points belong to class 1 & are denoted by plus sign.
- 1 point belongs to class 0 and is denoted by minus sign.

Boost



Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

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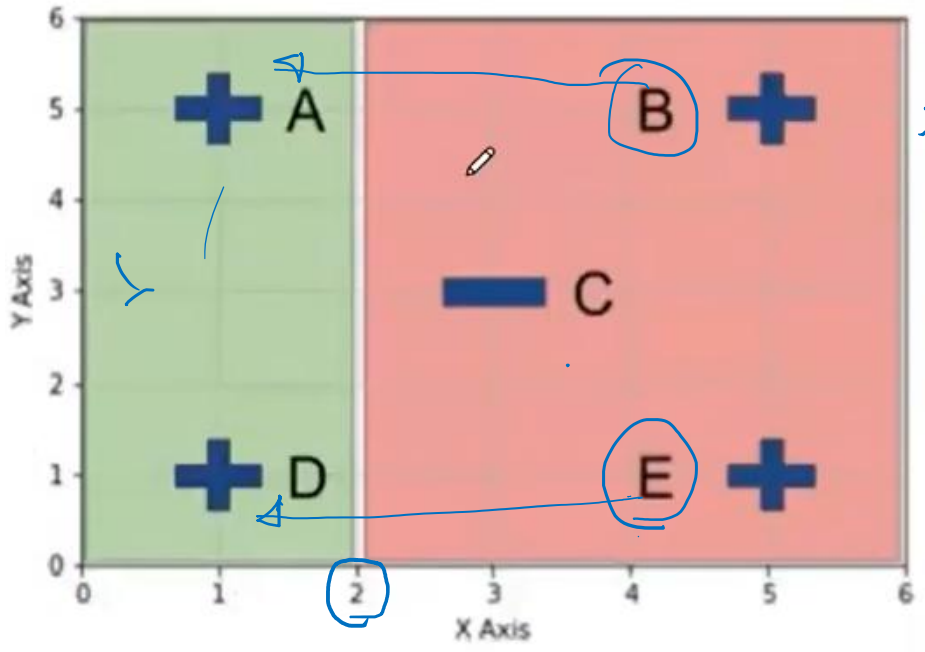

C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6		
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

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Pick Lowest Error Rate Classifier

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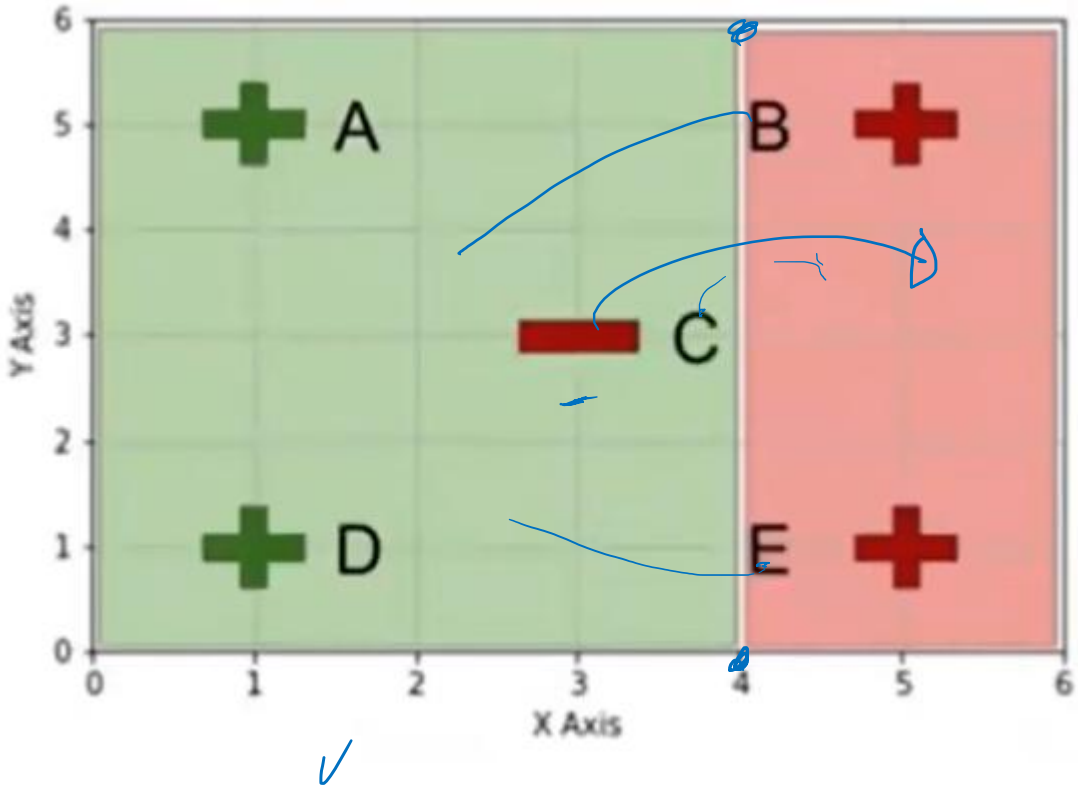

C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6		
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/5
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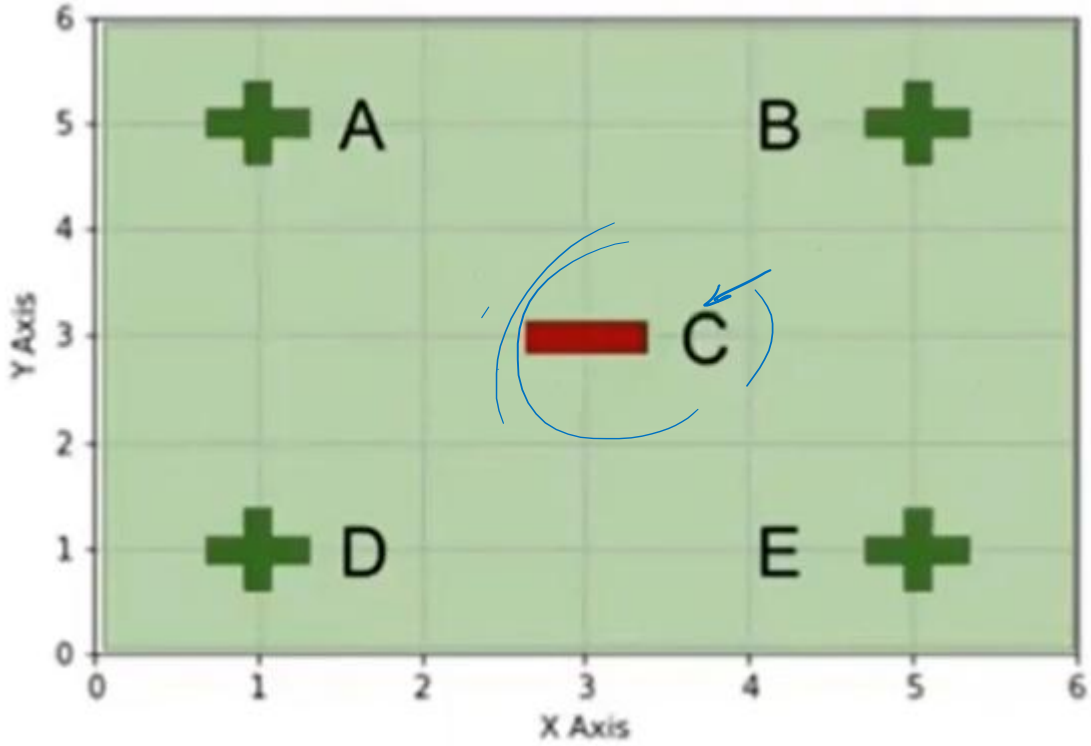

C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6	C	1/5
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

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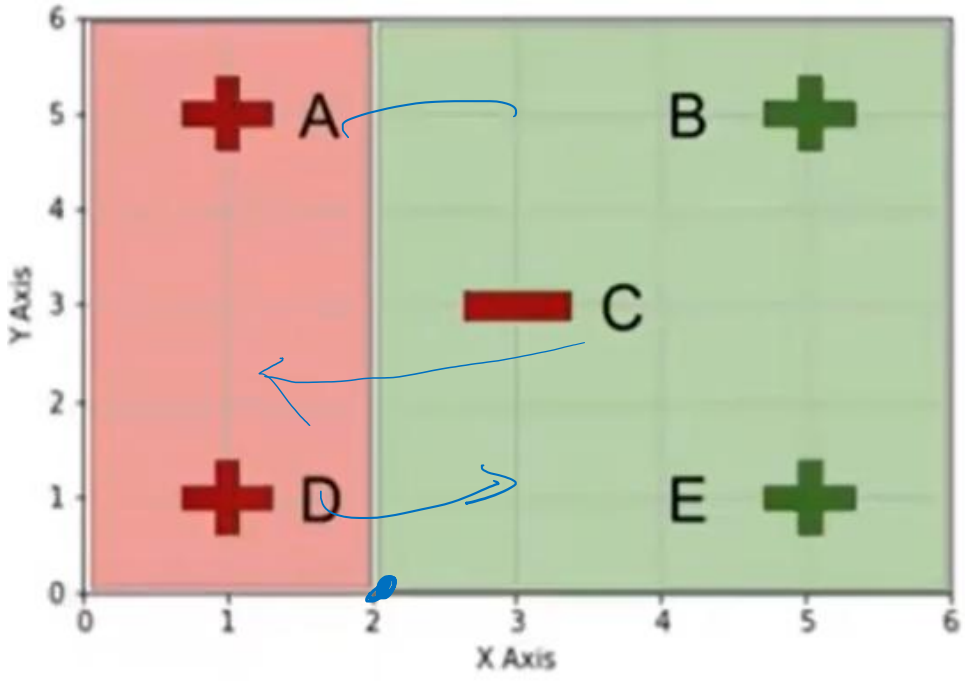

C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6	C	1/5
X > 2	A, D & C	3/5
X > 4		
X > 6		

Points	Weight
Wa	1/5
Wb	1/5
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Boost

$$w = \frac{1}{N}$$

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Pick Lowest Error Rate Classifier

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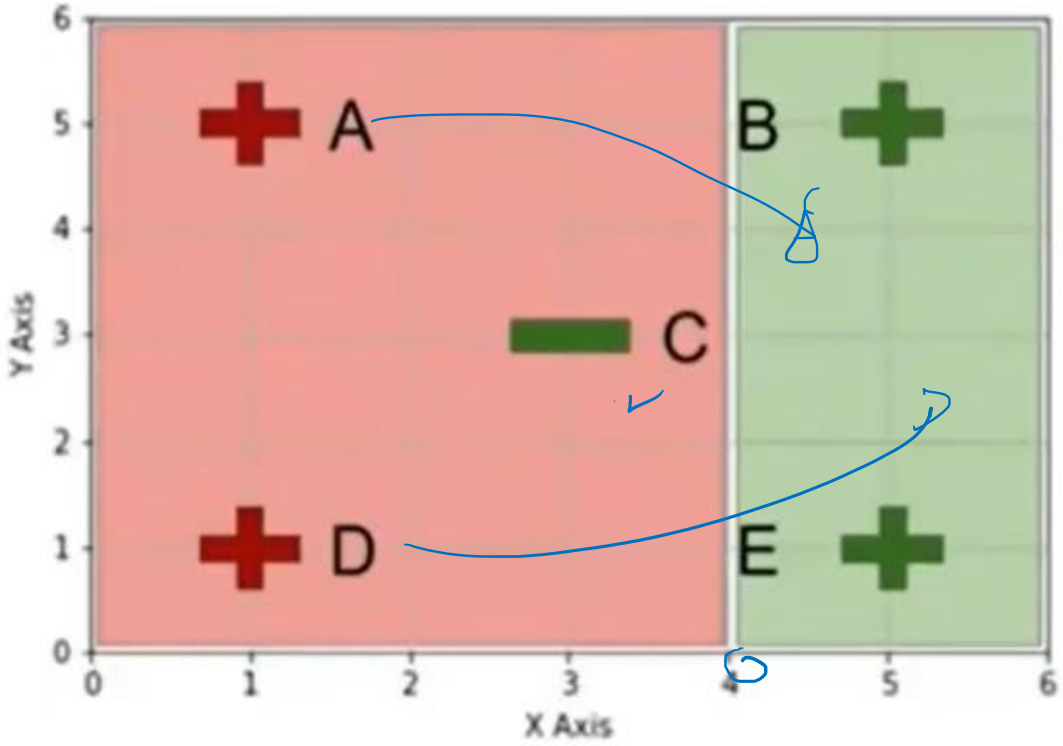

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X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6	C	1/5
X > 2	A, D & C	3/5
X > 4	A, D	2/5
X > 6		

Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

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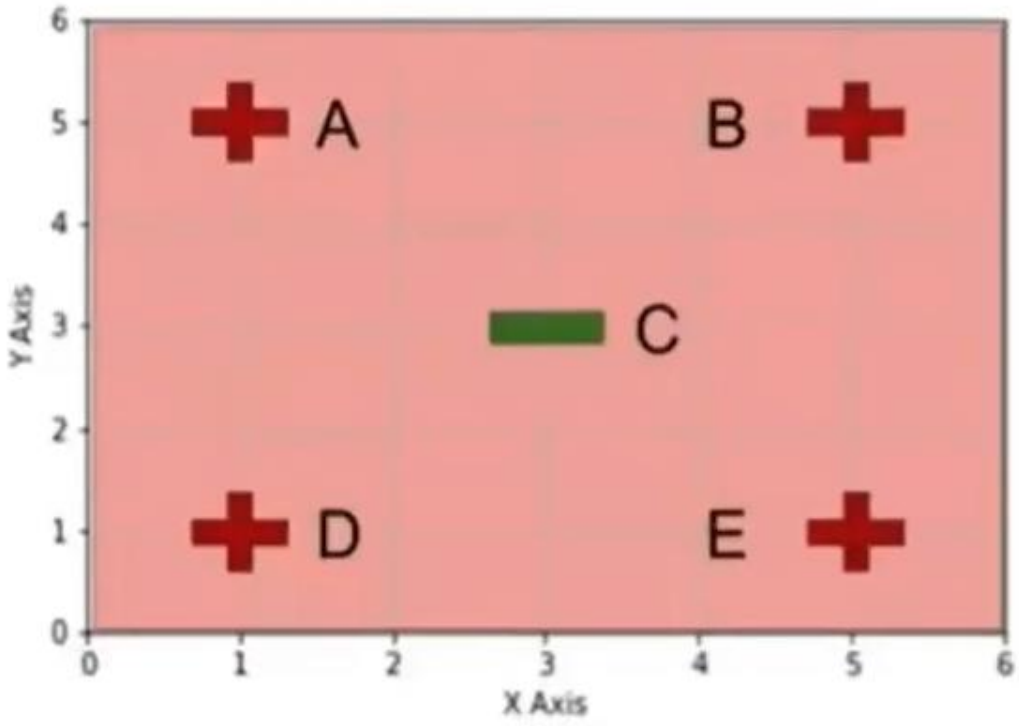


C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6	C	1/5
X > 2	A, D & C	3/5
X > 4	A, D	2/5
X > 6	A, B, D, E	4/5

Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$

C	Wrong	Error
X < 2	B & E	2/5
X < 4	B, C & E	3/5
X < 6	C	1/5
X > 2	A, D & C	3/5
X > 4	A, D	2/5
X > 6	A, B, D, E	4/5

- Going through the error rates of each classifier, we find that X< 6 is the best performing classifier.
- Now, that we have decided on the classifier, let's now find the voting power of the classifier.

Points	Weight
Wa	1/5
Wb	1/5
Wc	1/5
Wd	1/5
We	1/5

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$

$$\epsilon = 1/5$$

$$\alpha = \frac{1}{2}\log\frac{1-\frac{1}{5}}{\frac{1}{5}}$$

$$\alpha = \frac{1}{2}\log 4$$

$$h(x) = \frac{1}{2}\log 4 * F(x < 6)$$

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} & \text{if point classified correctly} \\ \frac{w_{old}}{2\epsilon} & \text{if point classified wrongly} \end{cases}$$

0.2 / 2*0.8 = 0.125
= 1/8

0.2/2*0.2=0.5
1/2

1/2

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
We	1/8

- Points A, B, D and E are correctly classifier.
- Plugging in the value of error = ⅕ and initial weight of ⅕ in the 1st equation.
- We get new weight = ⅛

Boost

$$w = \frac{1}{N}$$

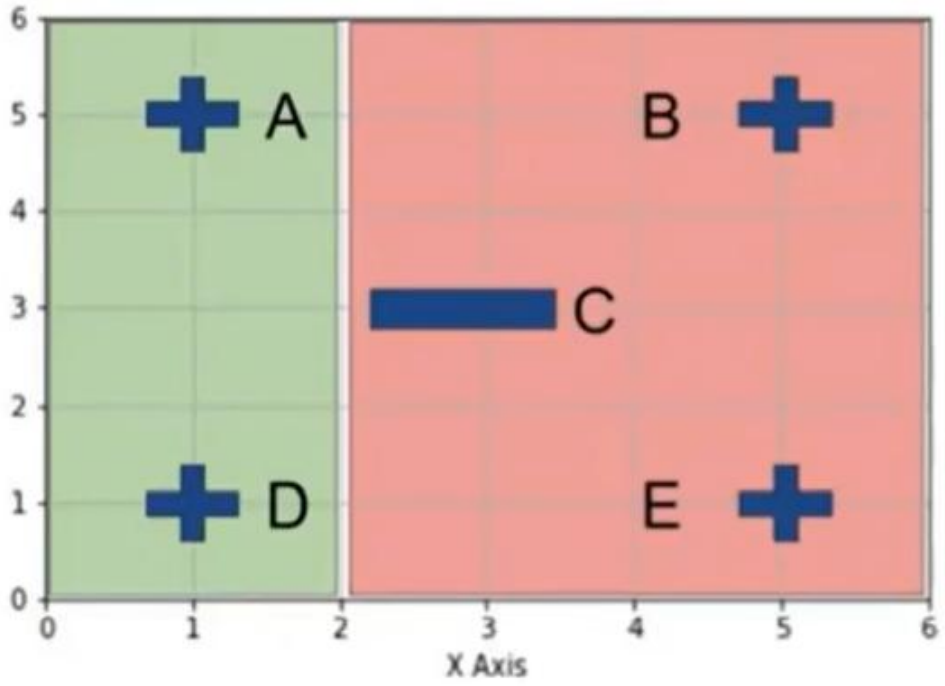
$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$



C	Wrong	Error
X < 2	B & E	2/8
X < 4		
X < 6		
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
We	1/8

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

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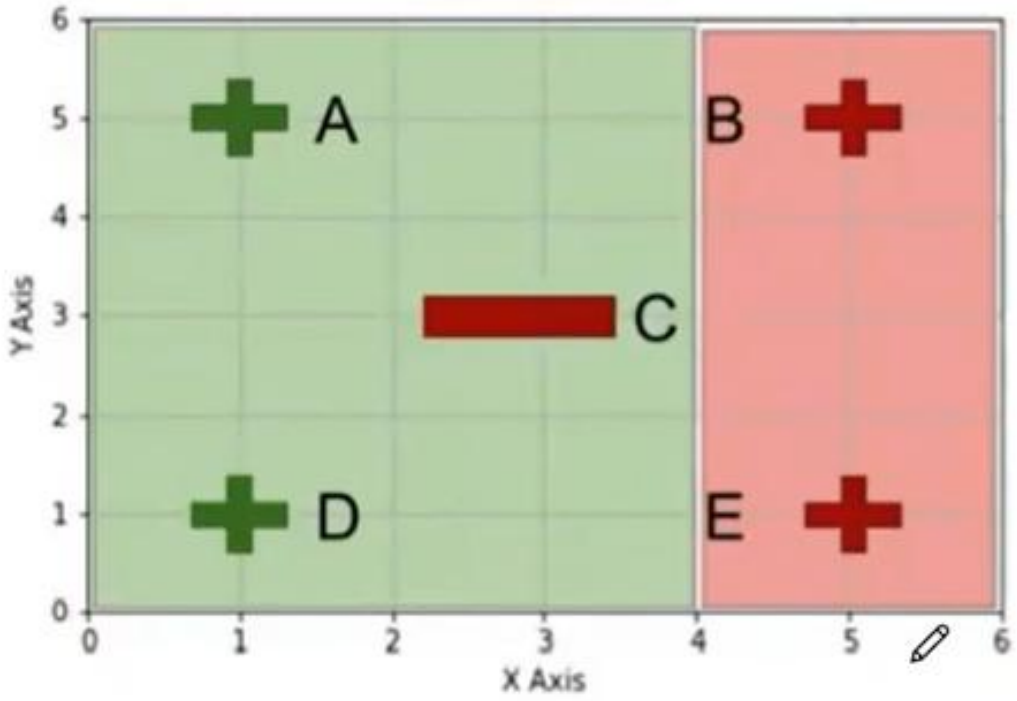

C	Wrong	Error
X < 2	B & E	2/8
X < 4	B, C & E	6/8
X < 6		
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
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Boost

$$w = \frac{1}{N}$$

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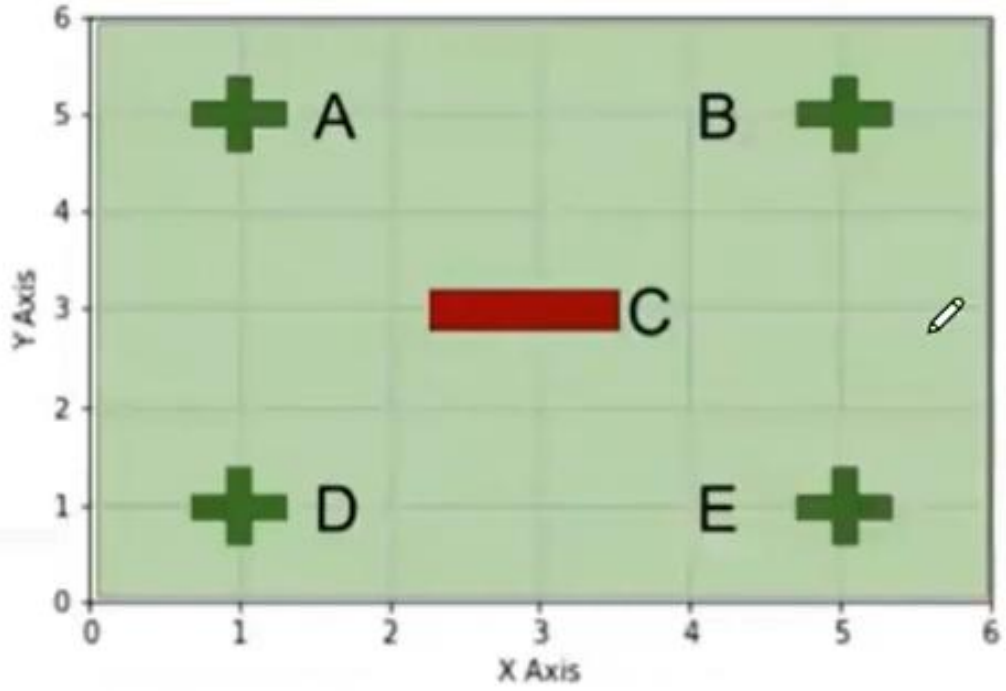

C	Wrong	Error
X < 2	B & E	2/8
X < 4	B, C & E	6/8
X < 6	C	4/8
X > 2		
X > 4		
X > 6		

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
We	1/8

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2} \log \frac{1-\epsilon}{\epsilon}$$

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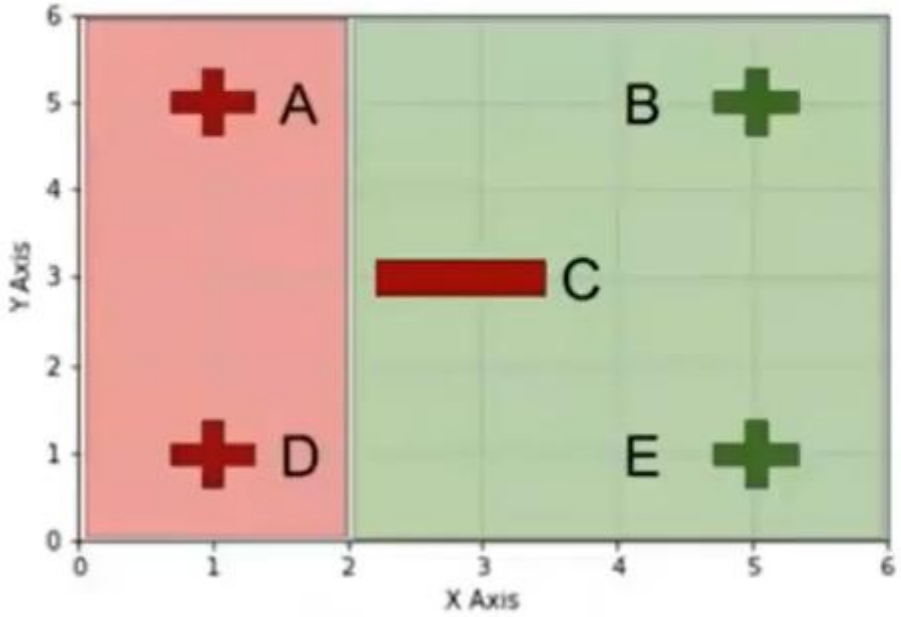

C	Wrong	Error
X < 2	B & E	2/8
X < 4	B, C & E	6/8
X < 6	C	4/8
X > 2	A, C & D	6/8
X > 4		
X > 6		

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
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Boost

$$w = \frac{1}{N}$$

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Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

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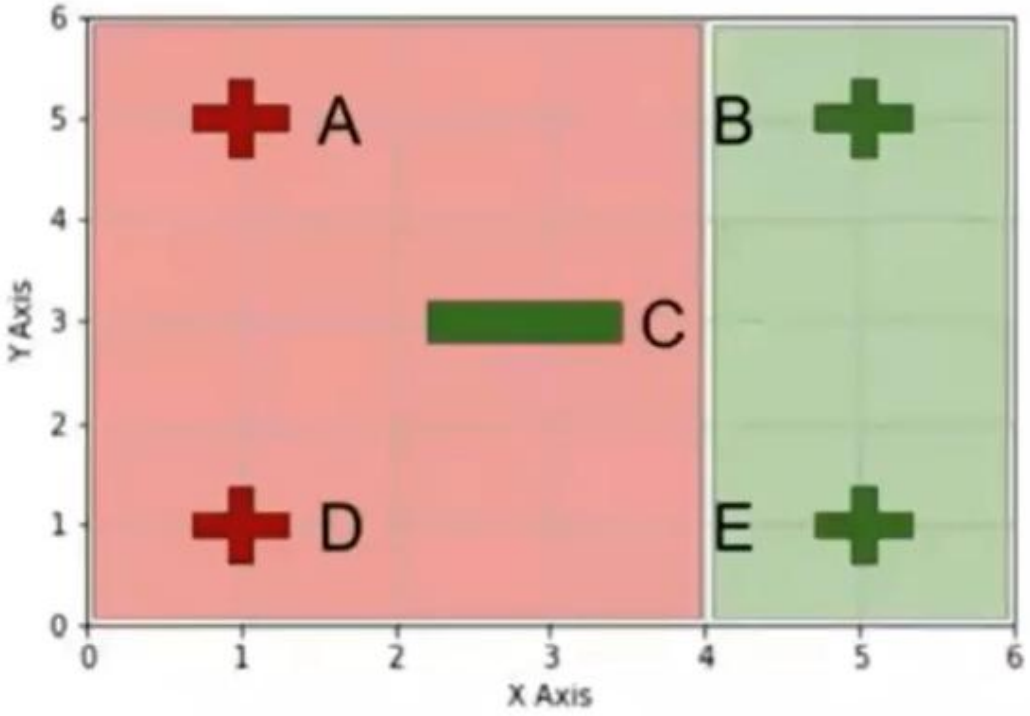

C	Wrong	Error
X < 2	B & E	2/8
X < 4	B, C & E	6/8
X < 6	C	4/8
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X > 4	A, D	2/8
X > 6		

Points	Weight
Wa	1/8
Wb	1/8
Wc	1/2
Wd	1/8
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Boost

$$w = \frac{1}{N}$$

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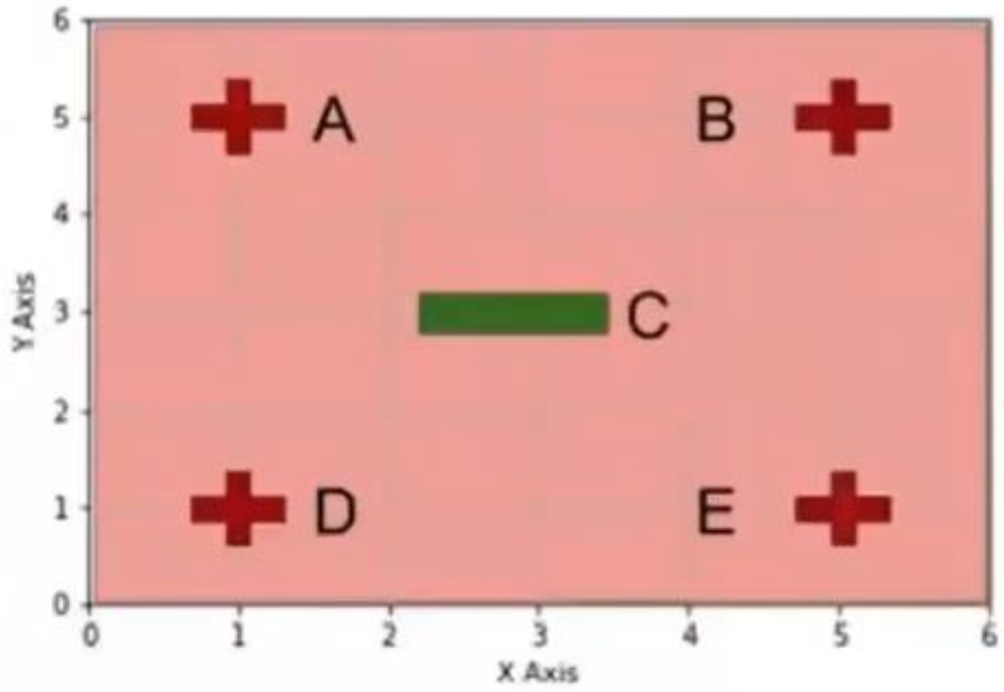

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X < 6	C	4/8
X > 2	A, C & D	6/8
X > 4	A, D	2/8
X > 6	A, B, D, E	4/8

Points	Weight
Wa	1/8
Wb	1/8
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Wd	1/8
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Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$

C	Wrong	Error
X < 2	B & E	2/8
X < 4	B, C & E	6/8
X < 6	C	4/8
X > 2	A, D & C	6/8
X > 4	A, D	2/8
X > 6	A, B, D, E	4/8

- Going through the error rates of each classifier, we find that there are 2 candidates X< 2 and X> 4 with minimum error rate.
- In this round, we will choose the 1st one i.e. X< 2 classifier.

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$

$$\epsilon = 2/8$$

$$\alpha = \frac{1}{2}\log\frac{1-\frac{1}{4}}{\frac{1}{4}}$$

$$\alpha = \frac{1}{2}\log 3$$

$$h(x) = \frac{1}{2}\log 4 * F(x < 6) + \frac{1}{2}\log 3 * F(x < 2)$$

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$a = \frac{1}{2}\log\frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} a_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} & \text{if point classified correctly} \\ \frac{w_{old}}{2\epsilon} & \text{if point classified wrongly} \end{cases}$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} & \text{if point classified correctly} \\ \frac{w_{old}}{2\epsilon} & \text{if point classified wrongly} \end{cases}$$

- Points B and E are incorrectly classified.
- Plugging in the value of error = 2/8 and initial weights from previous round we get the following points weight table

Points	Weight
Wa	1/12
Wb	3/12
Wc	4/12
Wd	1/12
We	3/12

Boost

$$w = \frac{1}{N}$$

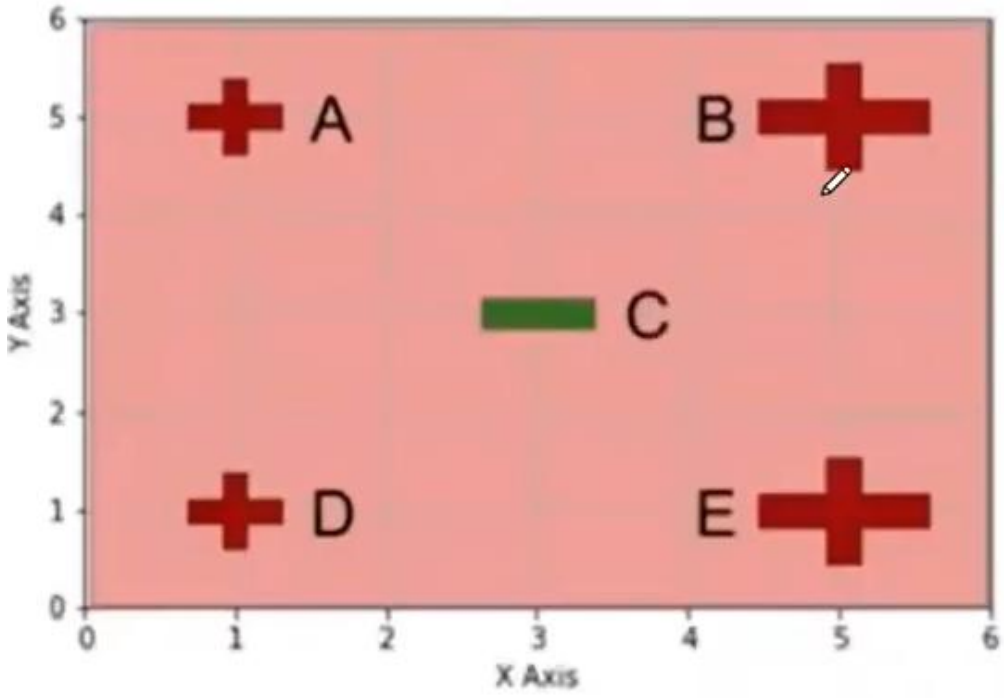
$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2} \log \frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$



C	Wrong	Error
X < 2	B & E	1/2
X < 4	B, C & E	10/12
X < 6	C	4/12
X > 2	A, C & D	1/2
X > 4	A, D	2/12
X > 6	A, B, D, E	8/12

Points	Weight
Wa	1/12
Wb	3/12
Wc	4/12
Wd	1/12
We	3/12

Boost

$$w = \frac{1}{N}$$

$$\epsilon = \sum_{\text{wrong}} w_i$$

Pick Lowest Error Rate Classifier

$$\alpha = \frac{1}{2} \log \frac{1-\epsilon}{\epsilon}$$

$$f(x_i) = \sum_{t=1}^{T} \alpha_t h_t(x_i)$$

$$w_{new} = \begin{cases} \frac{w_{old}}{2(1-\epsilon)} \\ \frac{w_{old}}{2\epsilon} \end{cases}$$

$$\epsilon = 1/6$$

$$\alpha = \frac{1}{2} \log \frac{1-\frac{1}{6}}{\frac{1}{6}}$$

$$\alpha = \frac{1}{2} \log 5$$

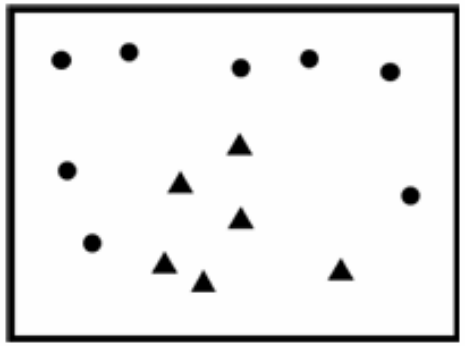
$$h(x) = \frac{1}{2} \log 4 * F(x < 6) + \frac{1}{2} \log 3 * F(x < 2) + \frac{1}{2} \log 5 * F(x > 4)$$

AdaBoost

- Stands for *Adaptive Boosting*
- Combines weak learners **linearly**
- Most commonly used to boost **decision trees** with one level, aka **decision stumps**
- Iteratively adapts to the errors made by weak learners in previous iterations.
- **Re-weighting** scheme:
  - Higher weight is assigned to incorrectly classified data points
  - Lower weight assigned to correctly classified data points

Initialisation

Training data



Initialise the **weight matrix** of data points as uniform distribution, i.e. assigning same weight to all data points:

$$\alpha_i = 1 / N$$

where N is the number of data points.

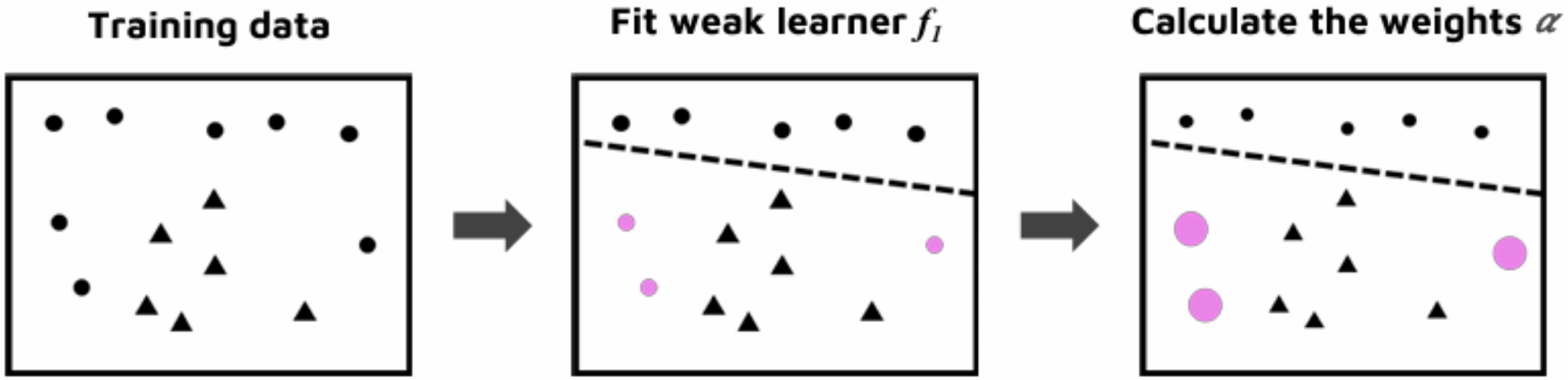
Iterations

For $i = 1, \ldots, T$ or until low enough error is achieved:

- Fit weak learner $f_t(x)$ to data points with data point weights $\boldsymbol{\alpha}_i$
- Compute weak learner's weight $\boldsymbol{w_t} = \frac{1}{2}\ln\left(\frac{1 - weighted_error(f_t)}{weighted_error(f_t)}\right)$
- Recompute data point weights $\boldsymbol{\alpha}_i = \begin{cases} \alpha_i e^{-w_t} & \text{if } f_t(x_i) = y_i \\ \alpha_i e^{w_t} & \text{if } f_t(x_i) \neq y_i \end{cases}$
- Normalize weights $\boldsymbol{\alpha}_i$

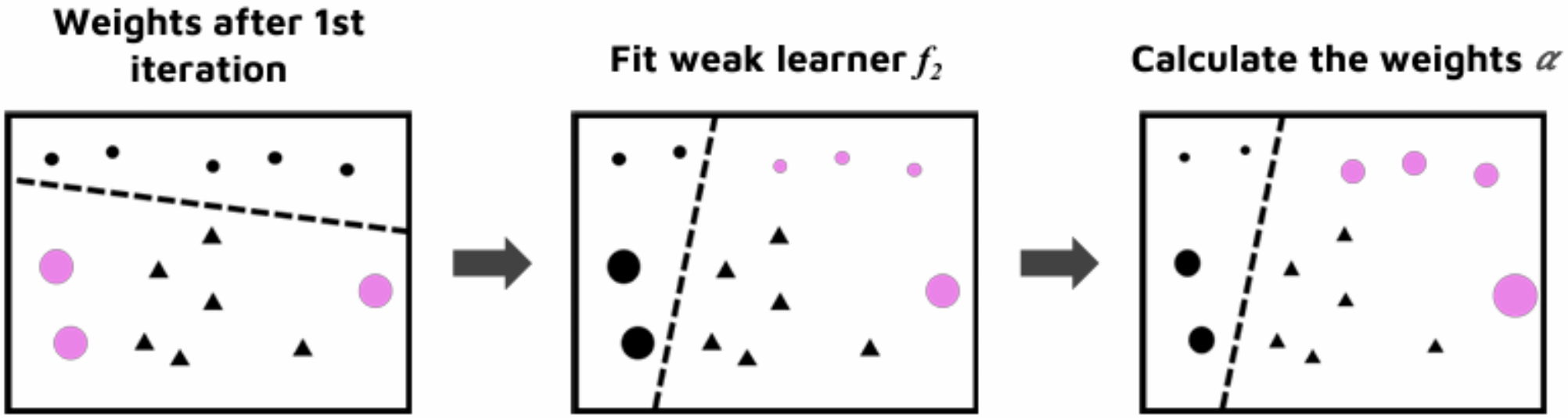
Boost

Iteration 1



- initially all data points have the same weight 1/N
- whatever the class is correctly classified will be given less weights in the next iteration, and higher weights for misclassified classes

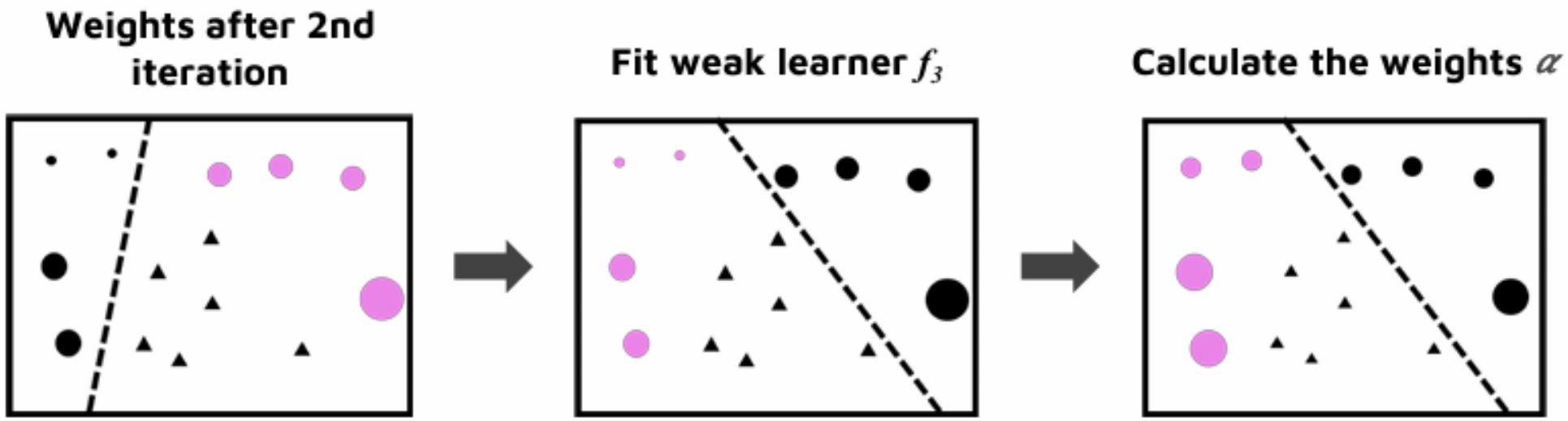
Iteration 2



- Weak learner forms a decision boundary which classifies better the data points with higher weights

Boost

Iteration 3



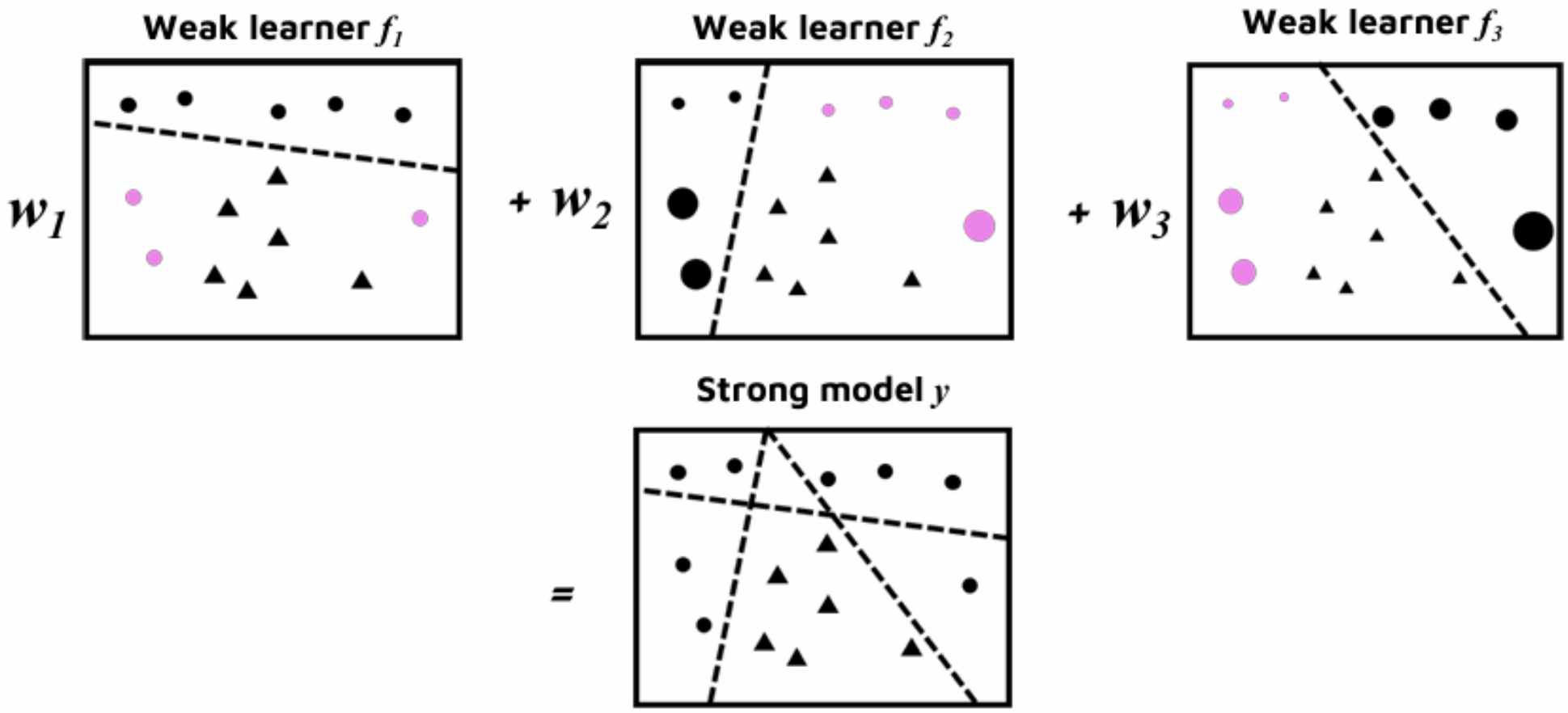
... continue iterating until either:

- Sufficiently low training error is achieved (with enough iterations the algorithm can reach 100% accuracy)
- A predefined number of weak learners was added

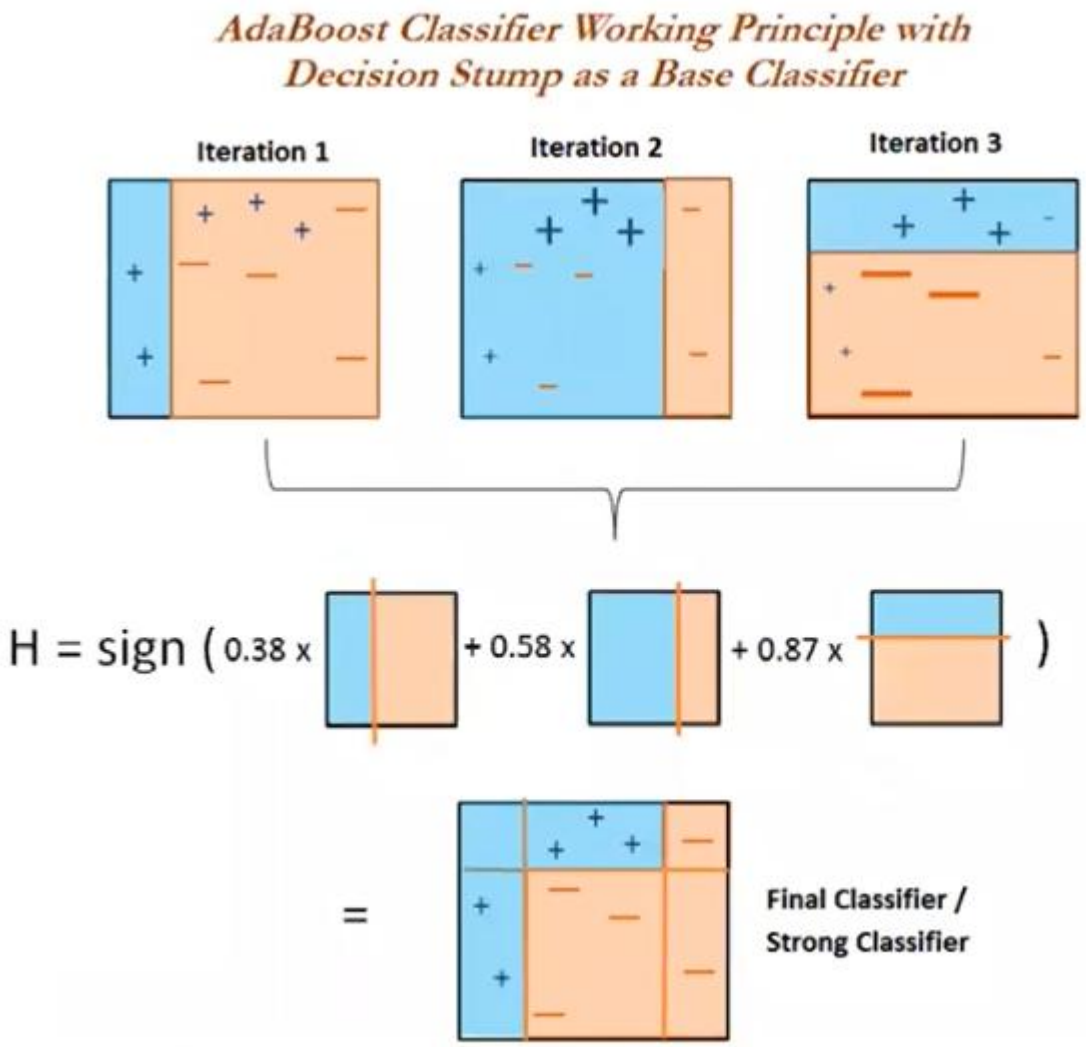
Boost

Final model

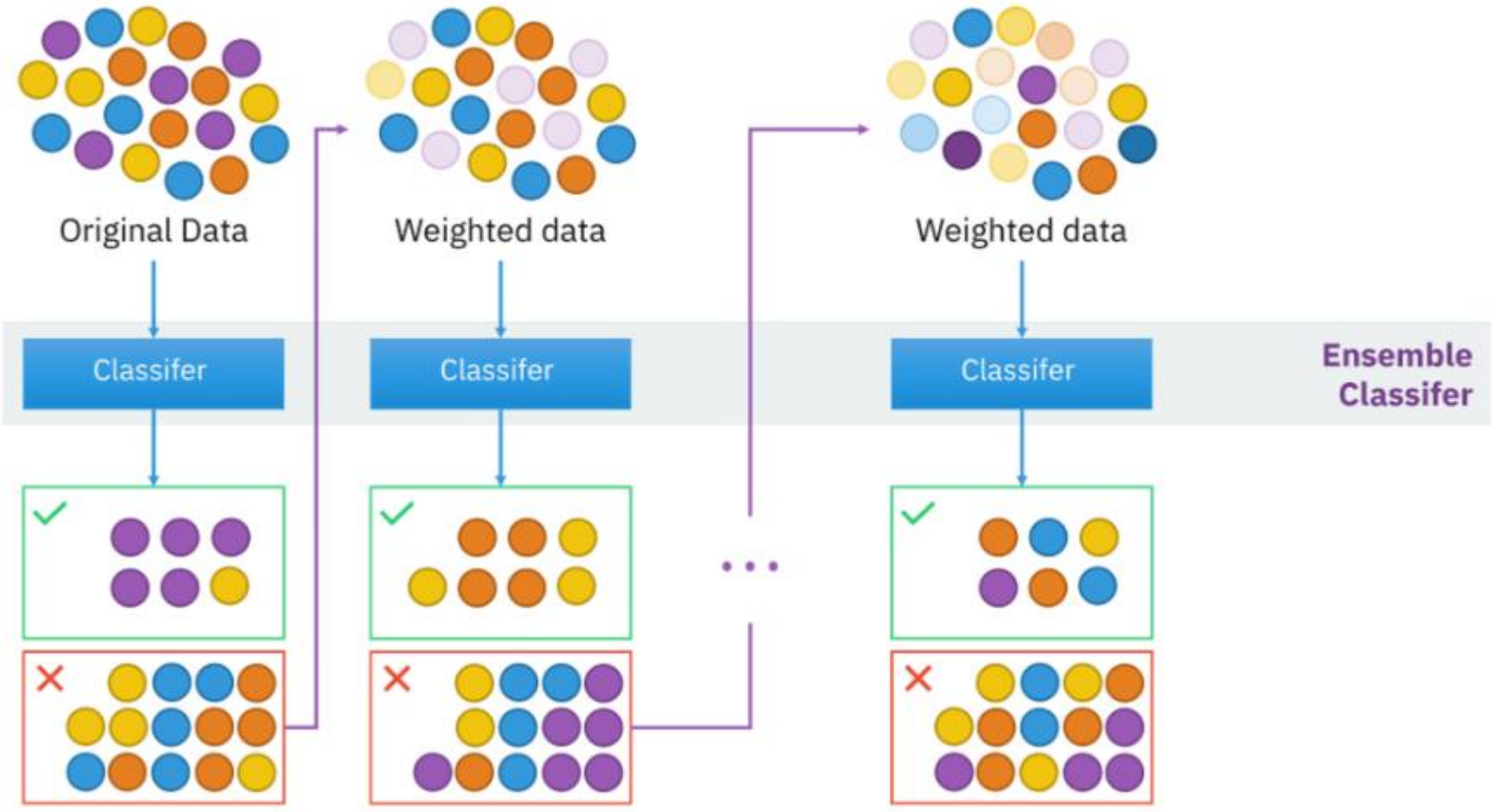
$$\hat{y} = sign\left(\sum_{t=1}^{T} w_t f_t(x)\right)$$



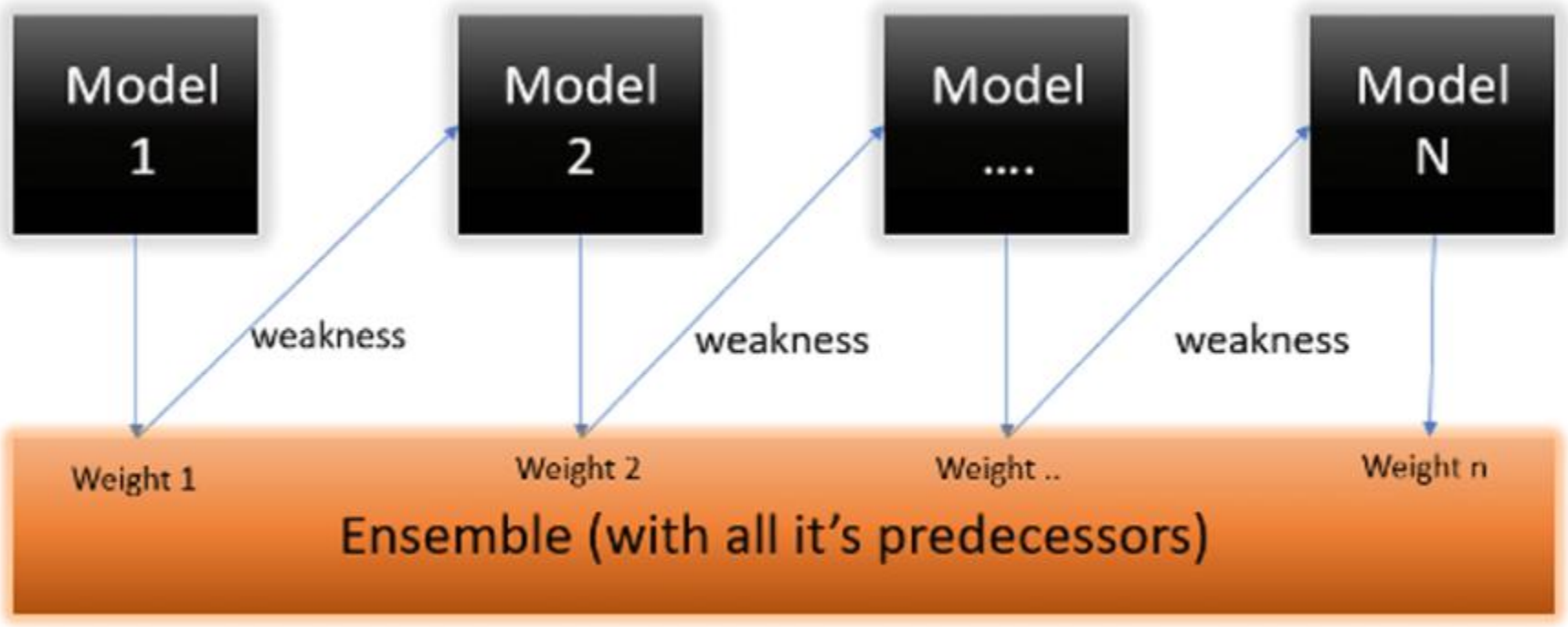
Boost


AdaBoost Classifier Working Principle with
Decision Stump as a Base Classifier
Iteration 1
Iteration 2
Iteration 3
H = sign (0.38 x
+ 0.58 x
+ 0.87 x
)
=
Final Classifier /
Strong Classifier

Boost


Original Data
Weighted data
Weighted data
Classifer
Classifer
Classifer
Ensemble Classifer

Boost


Model 1
Model 2
Model ....
Model N
weakness
weakness
weakness
Weight 1
Weight 2
Weight ..
Weight n
Ensemble (with all it's predecessors)

Boost

Step 1:

The image below represents the adaboost algorithm example or adaboost example by taking below dataset . It is a classification challenge since the target column is binary. First and foremost, these data points will be weighted. At first, all of the weights will be equal.

ROW NO.	GENDER	AGE	INCOME	ILLNESS	SAMPLE WEIGHTS
1	Male	41	40000	Yes	1/5
2	Male	54	30000	No	1/5
3	Female	42	25000	No	1/5
4	Female	40	60000	Yes	1/5
5	Male	46	50000	Yes	1/5

Boost

Step 2:

We will examine how well "Gender" classifies the samples, followed by how the variables (Age and Income) categorise the samples. We'll make a decision stump for each characteristic and then compute each tree's Gini Index. Our first stump will be the tree with the lowest Gini Index.

Let's suppose Gender has the lowest gini index in our dataset, thus it will be our first stump.

Boost

Step 3:

Using this approach, we will now determine the "Amount of Say" or "Importance" or "Influence" for this classifier in categorising the data points:

$$\frac{1}{2}\log\frac{1 - Total\ Error}{Total\ Error}$$

The total error is just the sum of all misclassified data points' sample weights.

If there is one incorrect output in our dataset, thus our total error is 1/5, and the alpha (performance of the stump) is:

Boost

$$Performance\ of\ the\ stump\ =\ \frac{1}{2}\log_e\left(\frac{1-Total\ Error}{Total\ Error}\right)$$

$$\alpha\ =\ \frac{1}{2}\log_e\left(\frac{1-\frac{1}{5}}{\frac{1}{5}}\right)$$

$$\alpha\ =\ \frac{1}{2}\log_e\left(\frac{0.8}{0.2}\right)$$

$$\alpha\ =\ \frac{1}{2}\log_e(4)\ =\ \frac{1}{2}*(1.38)$$

$$\alpha\ =\ 0.69$$

0 represents a flawless stump, while 1 represents a bad stump.

Boost

Step 4:

You're probably asking why it's required to determine a stump's TE and performance. The answer is simple: we need to update the weights since if the same weights are used in the next model, the result will be the same as it was in the previous model.

The weights of the incorrect forecasts will be increased, while the weights of the successful predictions will be dropped. When we create our next model after updating the weights, we will assign greater weight to the points with higher weights.

After determining the classifier's significance and total error, we must update the weights using the following formula:

$$New\ sample\ weight\ =\ old\ weight\ *\ e^{\pm Amount\ of\ say\ (\alpha)}$$

- When the sample is successfully identified, the amount of, say, (alpha) will be negative.
- When the sample is misclassified, the amount of (alpha) will be positive.
- There are four correctly categorised samples and one incorrectly classified sample. In this case, the sample weight of the datapoint is 1/5, and the quantity of say/performance of the Gender stump is 0.69.

$$New\ sample\ weight\ =\ \frac{1}{5}\ *\ \exp(-0.69)$$

$$New\ sample\ weight\ =\ 0.2\ *\ 0.502\ =\ 0.1004$$

Boost

The adjusted weights for incorrectly categorised samples will be:

ROW NO.	GENDER	AGE	INCOME	ILLNESS	SAMPLE WEIGHTS	NEW SAMPLE WEIGHTS
1	Male	41	40000	Yes	1/5	0.1004
2	Male	54	30000	No	1/5	0.1004
3	Female	42	25000	No	1/5	0.1004
4	**Female**	**40**	**60000**	**Yes**	**1/5**	**0.3988**
5	Male	46	50000	Yes	1/5	0.1004

We know that the entire sum of the sample weights must equal one, but if we add all of the new sample weights together, we get 0.8004. To get this amount equal to 1, we will normalise these weights by dividing all the weights by the entire sum of updated weights, which is 0.8004. Hence, we get this dataset after normalising the sample weights, and the sum is now equal to 1.

Boost

ROW NO.	GENDER	AGE	INCOME	ILLNESS	SAMPLE WEIGHTS	NEW SAMPLE WEIGHTS
1	Male	41	40000	Yes	1/5	0.1004/0.8004 = 0.1254
2	Male	54	30000	No	1/5	0.1004/0.8004 = 0.1254
3	Female	42	25000	No	1/5	0.1004/0.8004 = 0.1254
4	**Female**	**40**	**60000**	**Yes**	**1/5**	**0.3988/0.8004=0.4982**
5	Male	46	50000	Yes	1/5	0.1004/0.8004 = 0.1254

Boost

Step 5:

We must now create a fresh dataset to see whether or not the mistakes have decreased. To do this, we will delete the "sample weights" and "new sample weights" columns and then split our data points into buckets based on the "new sample weights."

ROW NO.	GENDER	AGE	INCOME	ILLNESS	SAMPLE WEIGHTS	NEW SAMPLE WEIGHTS	BUCKETS
1	Male	41	40000	Yes	1/5	0.1254	0 to 0.1254
2	Male	54	30000	No	1/5	0.1254	0.1254 to 0.2508
3	Female	42	25000	No	1/5	0.1254	0.2508 to 0.3762
4	**Female**	**40**	**60000**	**Yes**	**1/5**	**0.4982**	**0.3762 to 0.8744**
5	Male	46	50000	Yes	1/5	0.1254	0.8744 to 0.9998

Boost

Step 6:

We're nearly there. The method now chooses random values ranging from 0 to 1. Because improperly categorised records have greater sample weights, the likelihood of picking them is relatively high.

Assume the five random integers chosen by our algorithm are 0.38,0.26,0.98,0.40,0.55.

Now we'll examine where these random numbers go in the bucket and create our new dataset, which is displayed below.

ROW NO.	GENDER	AGE	INCOME	ILLNESS
1	Female	40	60000	Yes
2	Male	54	30000	No
3	Female	42	25000	No
4	Female	40	60000	Yes
5	Female	40	60000	Yes

This is our new dataset, and we can see that the data point that was incorrectly categorised has been picked three times since it has a greater weight.

Boost

Step 7:

This now serves as our new dataset, and we must repeat all of the preceding stages, i.e. Give each data point an equal weight. Determine the stump that best classifies the new group of samples by calculating their Gini index and picking the one with the lowest Gini index. To update the prior sample weights, compute the "Amount of Say" and "Total error." Normalize the newly calculated sample weights. Iterate through these procedures until a low training error is obtained.

Assume that we have built three decision trees (DT1, DT2, and DT3) sequentially with regard to our dataset. If we transmit our test data now, it will go through all of the decision trees, and we will eventually find which class has the majority, and we will make predictions for our test dataset based on that.

Boost

Quiz

1. What is the fundamental principle behind Adaboost?

1. Boosting
2. Bagging
3. Stacking
4. Ensembling

Boost

2.What is the main purpose of Adaboost?

1. To reduce bias and variance
2. To improve prediction accuracy
3. To reduce overfitting
4. To increase data complexity

Boost

3. What is the main difference between Adaboost and Bagging?

1. Bagging increases bias while Adaboost decreases bias
2. Bagging reduces variance while Adaboost increases variance
3. Adaboost is used for regression while Bagging is used for classification
4. Adaboost is used for classification while Bagging is used for regression

Boost

4.What is the main advantage of Adaboost over other ML algorithms?

1. It is more accurate
2. It is faster
3. It is more robust
4. It is easier to implement

Why use AdaBoost?

- Needs only a simple classifier as a weak learner
- Can achieve prediction results similar to powerful classifiers
- Can combine with any learning algorithm
- Requires little parameter tuning (usually only T)
- Selects only features known to improve predictive power
  - Relatively simple classifier, easy to program
  - Reduced dimensionality
  - Improved execution time
- Has been extended to problems beyond binary classification

However…

- Performance depends on the input data and weak learner
- Can fail if the weak classifiers are
  - Too complex (overfits)
  - Too weak (underfits)
- Sensitive to noisy data and outliers
- The ensemble is optimised based on the currently known estimates, not globally

Summary

- AdaBoost (i.e. Adaptive boosting) is one of the most popular and powerful **ensemble methods**
- Shifts algorithm's focus on the data points that are erroneous by **adapting weights**
- Simple to implement… depending on the weak learner you choose
- Vulnerable to noisy data

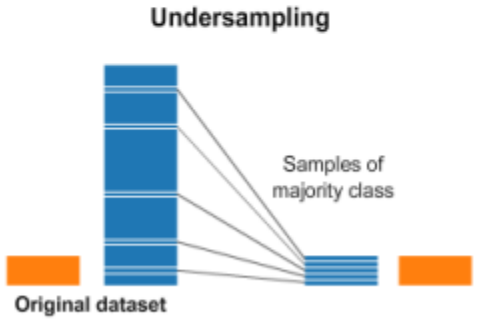
Imbalanced Classification

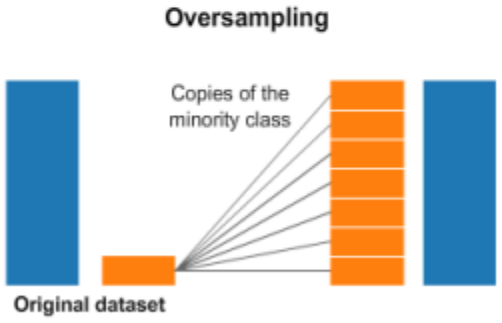
Imbalanced Classification:

- ➔ Imbalanced datasets are those where there is a severe skew in the classes distribution.
- ➔ This bias in the training dataset can influence many machine learning algorithms, leading some to ignore the minority class entirely.
- ➔ This is a problem as it is typically the minority class on which predictions are most important.

Imbalanced Data Pre-Processing :

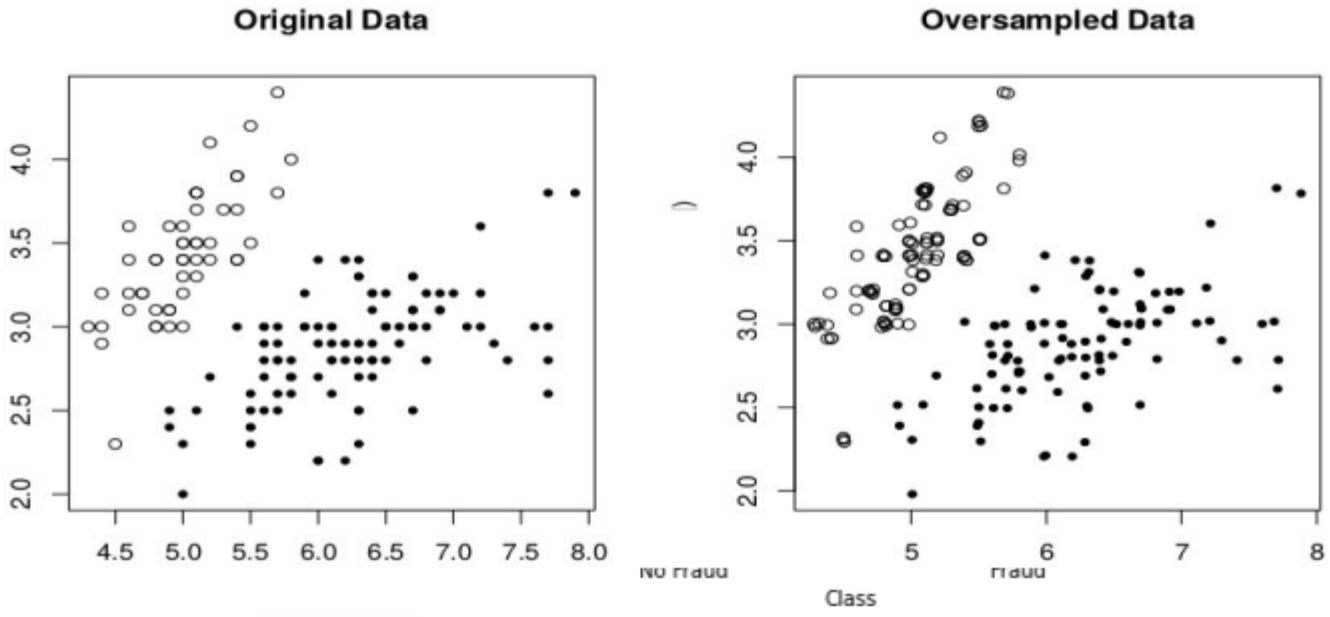
- Data Pre-Processing strategies are also known are Resampling Strategies - These are the most common strategies to tackle imbalanced domain learning task
- Over sampling:
  - Random Over sampling
  - SMOTE
- Under sampling:
  - Random Under sampling
  - Tomek links





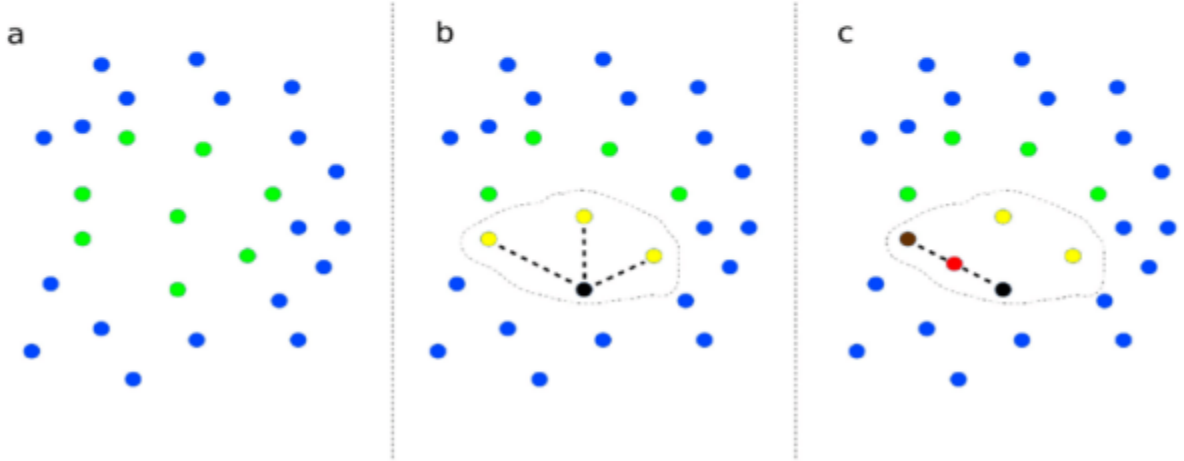
Random Over-sampling

➔ **Random oversampling** duplicates examples from the minority class in the training dataset and can result in overfitting for some models.



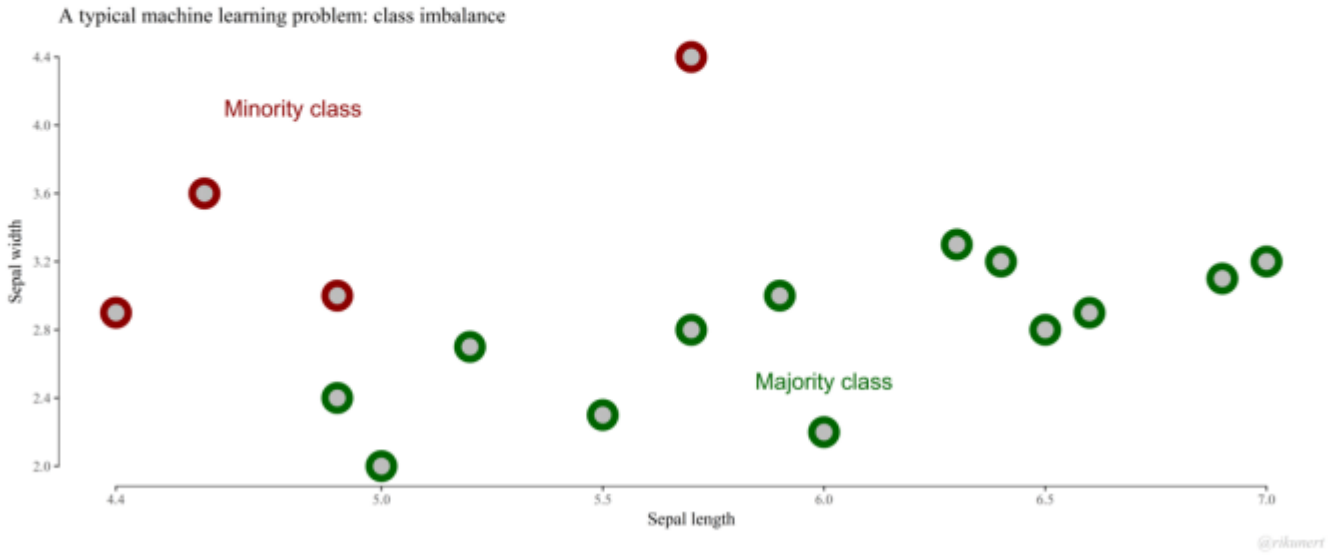
SMOTE

Synthetic Minority Oversampling Technique (SMOTE) works by selecting examples that are close in the feature space, drawing a line between the examples in the feature space and drawing a new sample at a point along that line.



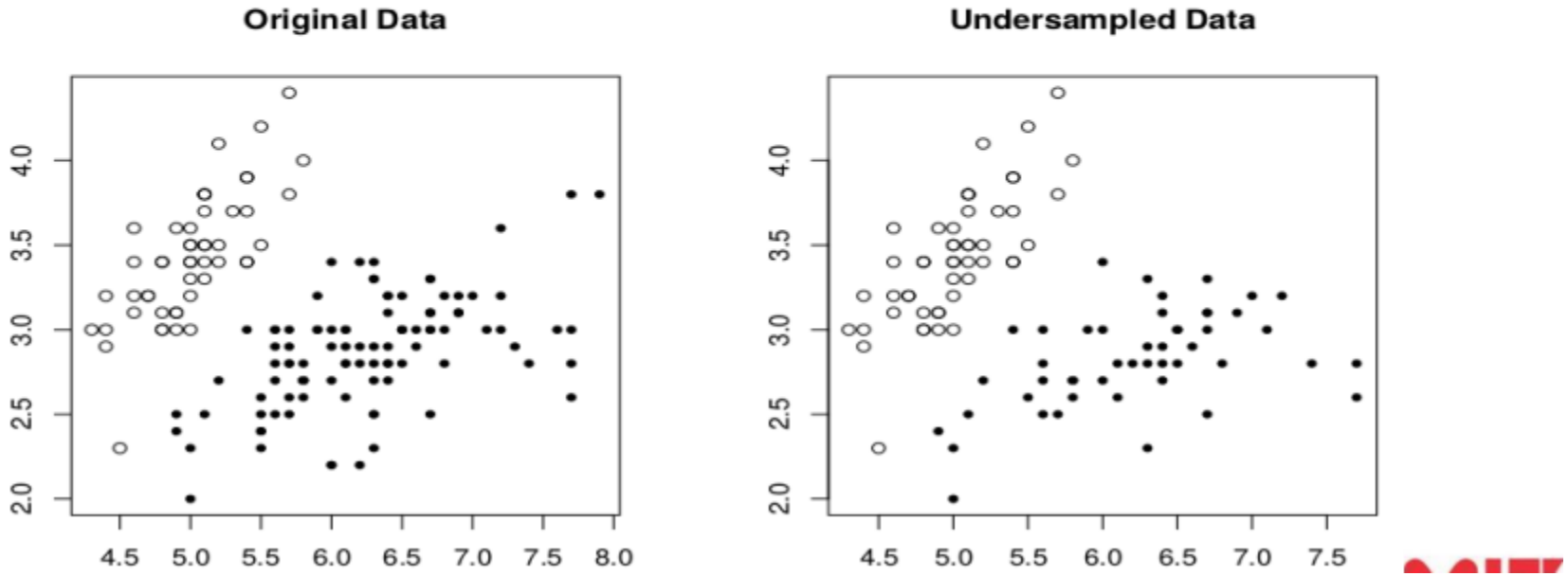
Boost

SMOTE


A typical machine learning problem: class imbalance
Minority class
Majority class
Sepal width
Sepal length
4.4
4.0
3.6
3.2
2.8
2.4
2.0
4.4
5.0
5.5
6.0
6.5
7.0

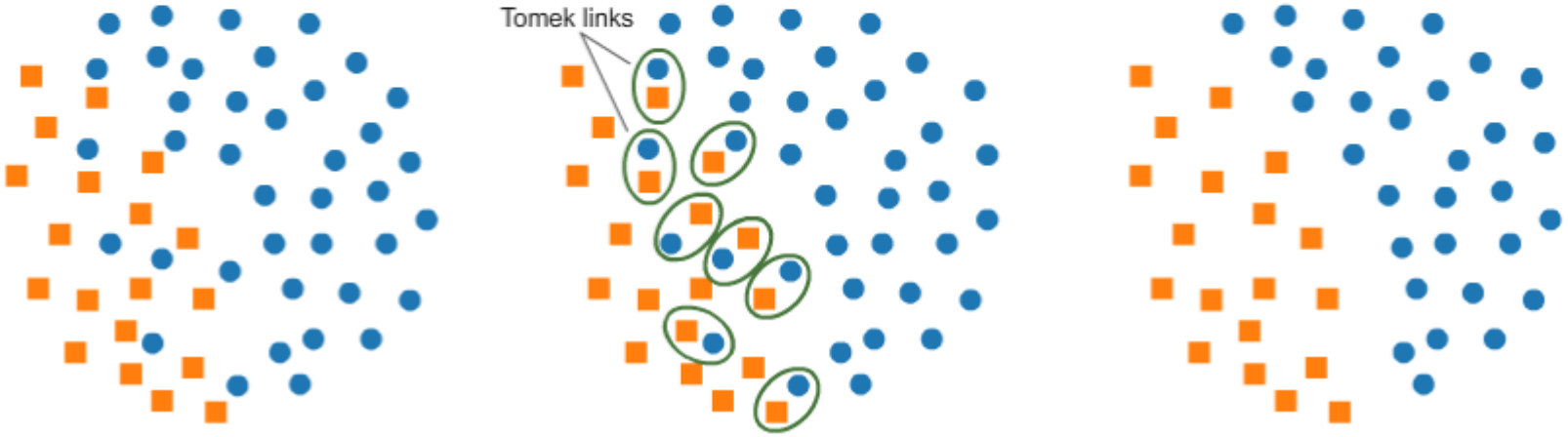
Random under - sampling

Random under – sampling Random under sampling deletes examples from the majority class and can result in losing information invaluable to a model.



Tomek links

Tomek's algorithm looks for such pairs and removes the majority instance of the pair. The idea is to clarify the border between the minority and majority classes, making the minority region(s) more distinct.



Imbalanced Classification:

Over Sampling:

➔ Random oversampling duplicates examples from the minority class in the training dataset and can result in overfitting for some models.

Under Sampling:

➔ Random undersampling deletes examples from the majority class and can result in losing information invaluable to a model.

Thank you